DETECTING ANOMALOUS ACTIVITY OF A SHIP ENGINE

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➢ Problem Overview

A shipping supply company aims to detect anomalous engine functionality, which represents unusual behavior that may affect ship engine performance. The dataset contains six engine-related features, each representing an indicator of engine condition and performance.

An anomaly is defined as an unusual combination of feature values across these engine features. Such anomalies may pose significant risks, including increased fuel consumption, reduced efficiency, delays in deliveries, and negative impact on revenue and customer satisfaction.

To address this problem, this project aims to develop an anomaly detection system capable of identifying potential engine problems at an early stage. Detecting these issues in advance allows the company to take preventive actions before serious damage occurs.

The approach combines exploratory data analysis, statistical techniques, and machine learning models to identify abnormal patterns in engine behavior. By applying these methods, the company can detect malfunctioning ship engines, reduce maintenance costs, minimize downtime, and improve operational safety.

➢ Approach

The analysis started with exploratory data analysis to understand the distribution of the six engine performance features. A statistical summary table was generated showing the count, mean, standard deviation, minimum, maximum, and quartiles for each feature [Figure 1]. This table helped identify the normal operating ranges of engine rpm, oil pressure, fuel pressure, coolant pressure, oil temperature, and coolant temperature.

From the statistical summary [Figure 1], it was observed that most values lie within a consistent range, while some maximum values are much higher than the upper quartile, suggesting the presence of possible outliers. These unusual values indicate that some ships may operate under abnormal engine conditions, which makes anomaly detection necessary.

After the initial statistical analysis, the data was scaled to ensure all features contributed equally to the models. Principal Component Analysis (PCA) was then applied to reduce the dataset to two dimensions for visualization purposes. This allowed the main cluster of normal observations and potential anomalies to be clearly seen.

Two anomaly detection methods were applied: One-Class SVM and Isolation Forest. Different parameters were tested, and the best results were obtained using Isolation Forest with contamination = 0.01 and 100 estimators, which produced an anomaly rate within the required range. The detected anomalies were plotted on the PCA projection to confirm that abnormal points appear outside the main cluster.

Histograms and boxplots [Figure 2] were also used to study the distribution of engine features. Most values lie within a normal range, forming dense clusters, while several outliers appear in the boxplots, especially for engine rpm and oil pressure. These extreme values indicate possible abnormal engine behavior and justify the use of anomaly detection methods.

	count	mean	std	min	25%	50%	75%	max
Engine rpm	19535.0	791.239263	267.611193	61.000000	593.000000	746.000000	934.000000	2239.000000
Lub oil pressure	19535.0	3.303775	1.021643	0.003384	2.518815	3.162035	4.055272	7.265566
Fuel pressure	19535.0	6.655615	2.761021	0.003187	4.916886	6.201720	7.744973	21.138326
Coolant pressure	19535.0	2.335369	1.036382	0.002483	1.600466	2.166883	2.848840	7.478505
lub oil temp	19535.0	77.643420	3.110984	71.321974	75.725990	76.817350	78.071691	89.580796
Coolant temp	19535.0	78.427433	6.206749	61.673325	73.895421	78.346662	82.915411	195.527912

Figure 1

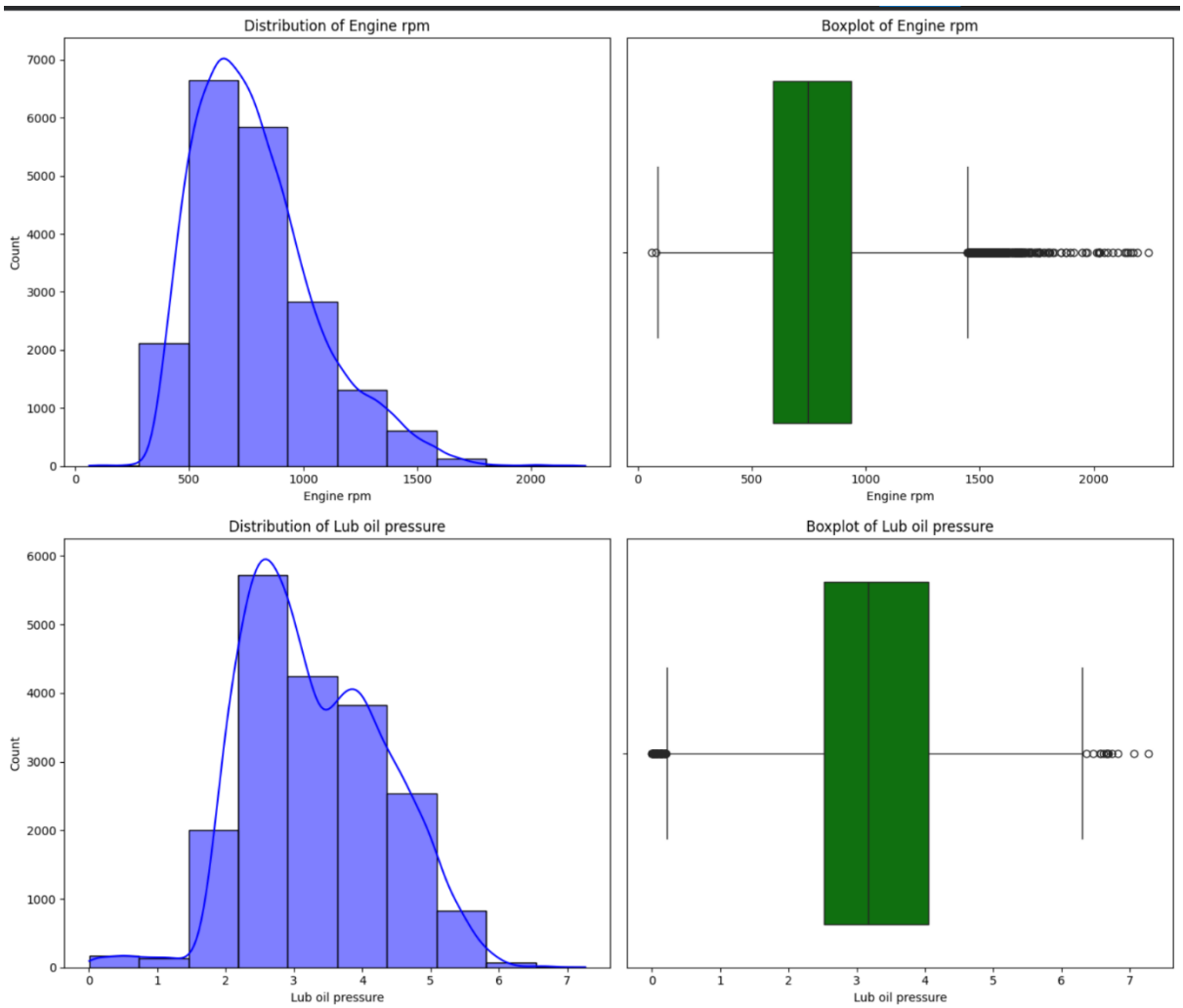

Figure 2

➢ Key Results

◆ IQR-Based Outlier Detection

The IQR method was applied to each feature, creating binary columns where 1 indicates an outlier. Features like lubricating oil temperature (13.4%) and fuel pressure (5.81%) had the most outliers, while coolant temperature (0.01%) and oil pressure (0.34%) were mostly stable [Figure 3].

By varying the number of features that must simultaneously be outliers, thresholds were set to control total anomalies. Threshold 1 flagged 23.7% of samples, threshold 2 flagged 2.16%, and stricter thresholds (3–6) flagged almost none. Using threshold 2 ensures that the percentage of detected anomalies falls within the desired 1–5% range, highlighting the most significant abnormal samples.

```
Engine rpm outliers: 464
Percentage: 2.375223957000256


Lub oil pressure outliers: 66
Percentage: 0.3378551318146916


Fuel pressure outliers: 1135
Percentage: 5.810084463782954


Coolant pressure outliers: 785
Percentage: 4.018428461735347


lub oil temp outliers: 2617
Percentage: 13.396467878167392


Coolant temp outliers: 2
Percentage: 0.010238034297414897


threshold 1 -> 23.73%
threshold 2 -> 2.16%
threshold 3 -> 0.06%
threshold 4 -> 0.00%
threshold 5 -> 0.00%
threshold 6 -> 0.00%
```

Figure 3

◆ Anomaly Detection Using One-Class SVM and PCA

To detect anomalies in high-dimensional feature data, a One-Class SVM with an RBF kernel was applied, with gamma set to "scale" and the nu parameter tuned to produce an anomaly rate within the required 1–5% range. Before modeling, the data was scaled, and PCA was applied separately for visualization purposes.

The 2D PCA plot [Figure 4] shows most points clustered together, with anomalies detected by SVM clearly separated from normal behavior. After testing multiple values, nu = 0.01 was selected, flagging approximately 1.01% of samples as anomalies. This ensures that the model focuses on the most significant deviations while avoiding overestimation of outliers.

Combining PCA and SVM allows both automated detection of unusual patterns and intuitive visualization of anomalous points.

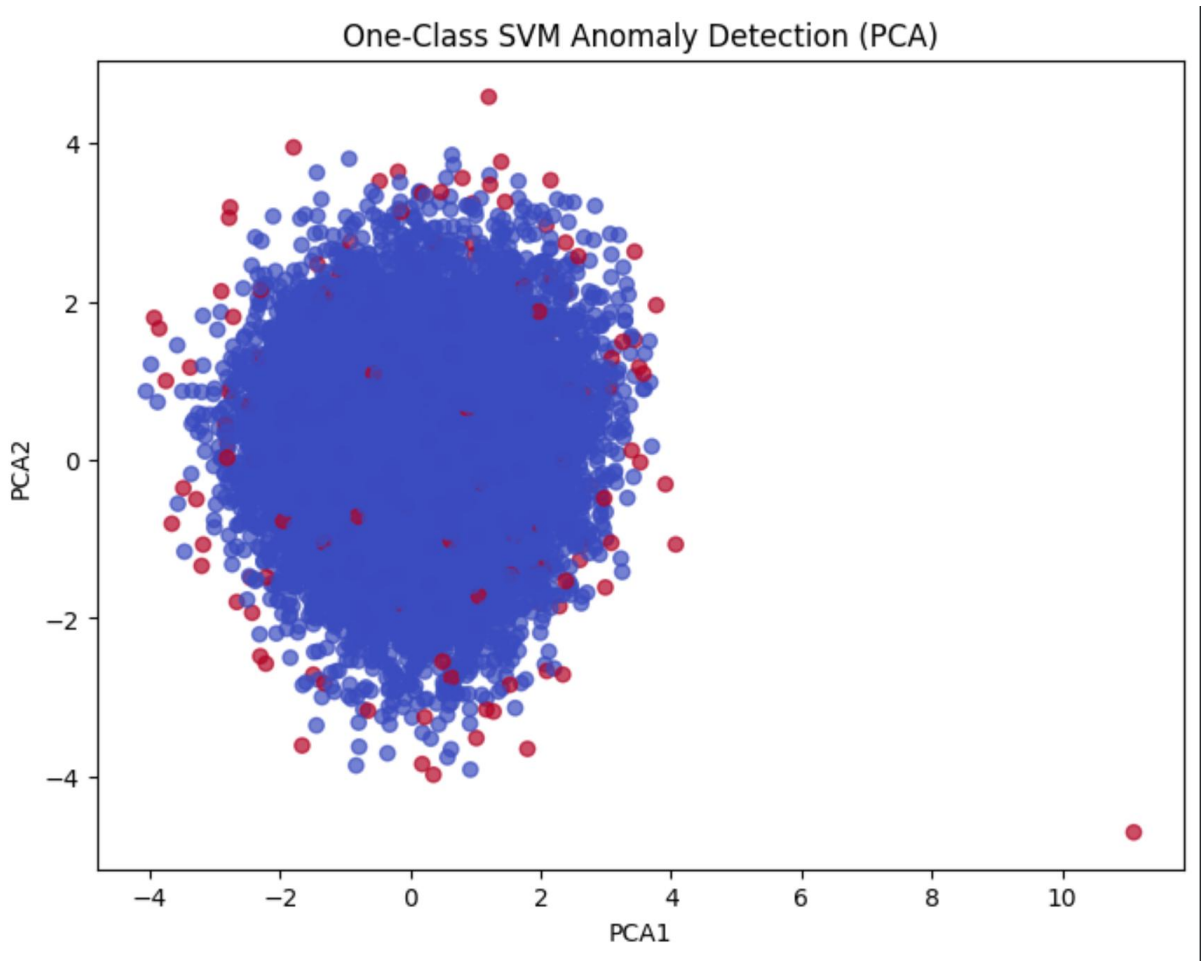

Figure 4

- **Anomaly Detection Using Isolation Forest and PCA**

  Isolation Forest was applied to detect anomalies in the feature data, with PCA reducing the data to two components for visualization. This method isolates points that deviate from the majority, effectively identifying unusual observations.

  The 2D PCA plot [Figure 5] shows normal points clustered together, while anomalies detected by Isolation Forest are clearly highlighted. The contamination parameter, controlling the expected fraction of anomalies, was tuned to achieve the target 1–5% range. The best value, contamination = 0.01, flagged approximately 1.00% of samples as anomalies, focusing on the most significant deviations without overestimating outliers.

  Combining PCA with Isolation Forest provides robust automated detection together with intuitive visual interpretation, complementing the One-Class SVM results.

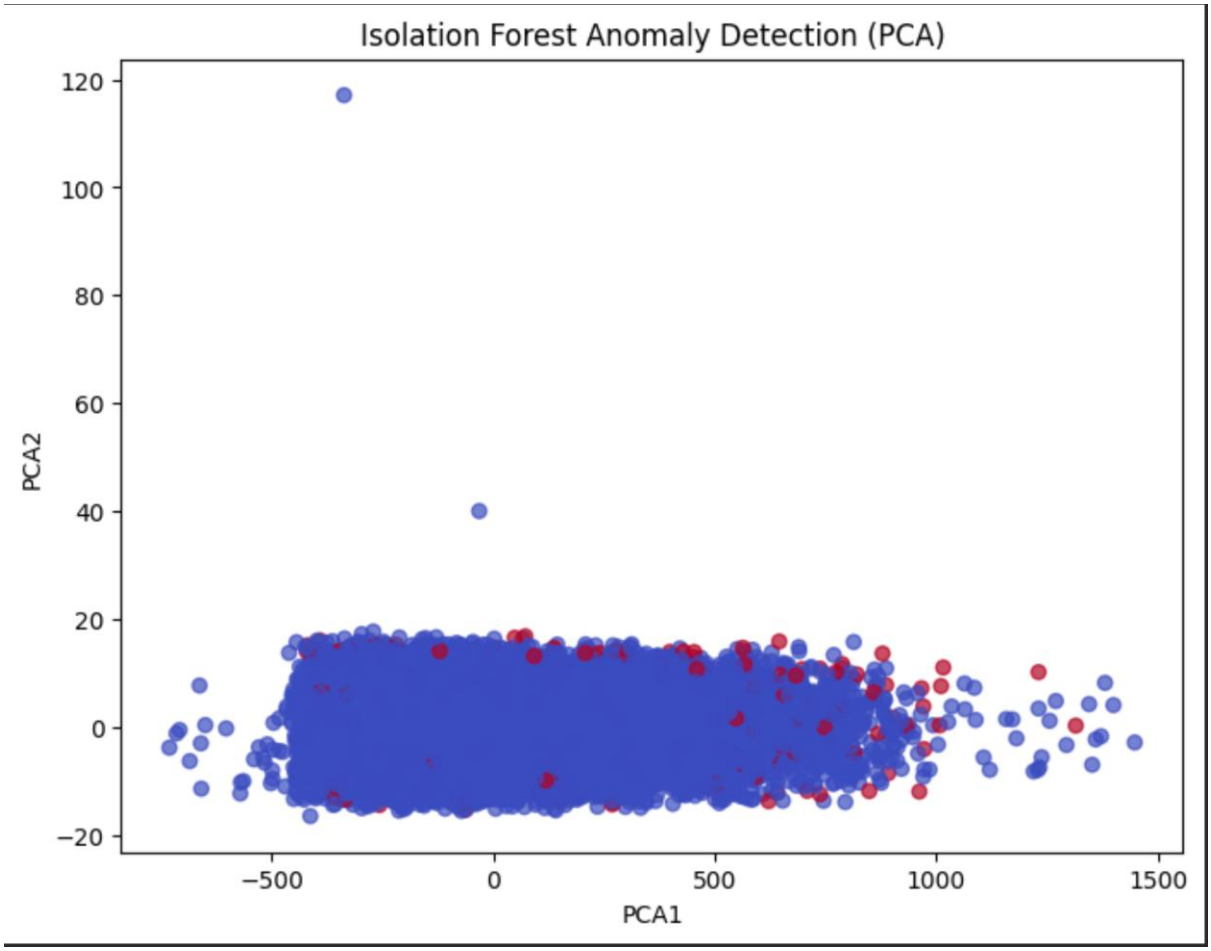

Figure 5

➢ Comparison of SVM and Isolation Forest with PCA

Both One-Class SVM and Isolation Forest were applied to the scaled sensor data, with PCA reducing the data to two components for visualization. The 2D PCA plots show that SVM detected anomalies mostly at the edges of the dense cluster, while Isolation Forest highlighted anomalies that are more isolated from the main distribution.

Parameter tuning ensured that both methods flagged approximately 1% of samples as anomalies (nu = 0.01 for SVM and contamination = 0.01 for Isolation Forest), keeping detection within the desired 1–5% range. Overall, SVM provides tighter boundary detection around the dense cluster, whereas Isolation Forest is more sensitive to extreme isolated points. Combining both methods provides a more reliable view of unusual engine behavior.

➢ Interpretation of the Anomaly Detection Results

The anomaly detection methods consistently identified a small proportion of samples as abnormal, indicating that most engine operations fall within normal ranges. The detected anomalies tend to appear at the edges of the PCA projection, suggesting that unusual engine conditions arise when multiple features deviate simultaneously.

The agreement between the IQR method, One-Class SVM, and Isolation Forest suggests that the detected anomalies are not random noise, but rather meaningful deviations in engine behavior. This consistency confirms that combining statistical and machine learning approaches improves the reliability of anomaly detection.

Although PCA visualization is useful for identifying general clustering patterns, reducing the data to two dimensions may lead to information loss, meaning some anomalies may not be fully separated.

➢ Recommendations

Based on the results, it is recommended to monitor engine features continuously using automated anomaly detection methods. Both Isolation Forest and One-Class SVM produced reliable results within the desired anomaly range and successfully identified abnormal engine conditions.

For future improvements, additional sensor features and nonlinear dimensionality reduction methods could be used to enhance detection accuracy. Implementing such systems in real time would allow early detection of engine problems, reduce maintenance costs, prevent failures, and improve overall operational safety.